

Sampling, bias, confidence intervals, precision

BEFORE-CLASS MIND MAP

Quantitative Methods for Business Analysis
About 10 minutes · read before class

HOW TO USE IT

1

Start with the big picture on page 2: how the 3 sections fit together.

2

Open each section map. Read each idea and its short note, and follow the labelled lines.

3

Answer the before-class question at the foot of each map and bring your answers to class.

INSIDE

S05

The big picture

How do we draw valid conclusions about business populations from sample data? **p. 2**

5.1

Sampling Design, Bias, and the Limits of Generalization

How do we draw a sample that truly represents the target population? **p. 3**

5.2

Sampling Distributions, Standard Error, and Mean Estimation

How precisely does a sample average estimate the true population mean? **p. 4**

5.3

Proportion Estimation, Sample Size Planning, and Decision Guardrails

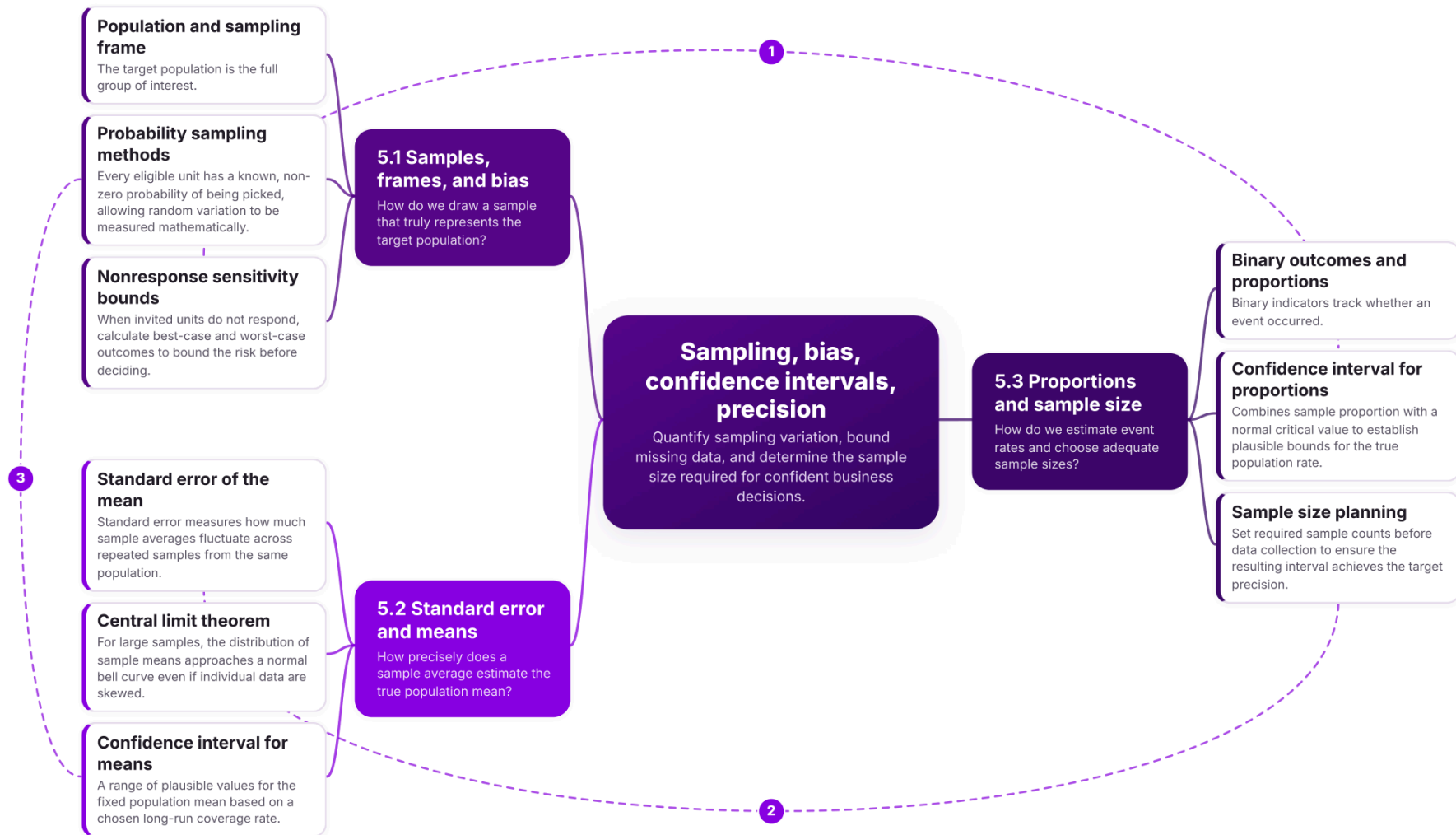
How do we estimate event rates and choose adequate sample sizes? **p. 5**

READING THE MAP

-  Idea
-  Idea written as a formula
-  Watch for this in class
-  Everyday example

Bring from earlier sessions: Expected value and uncertainty (Session 4). Explore how sampling design, standard error, and confidence intervals govern operational decisions under uncertainty.

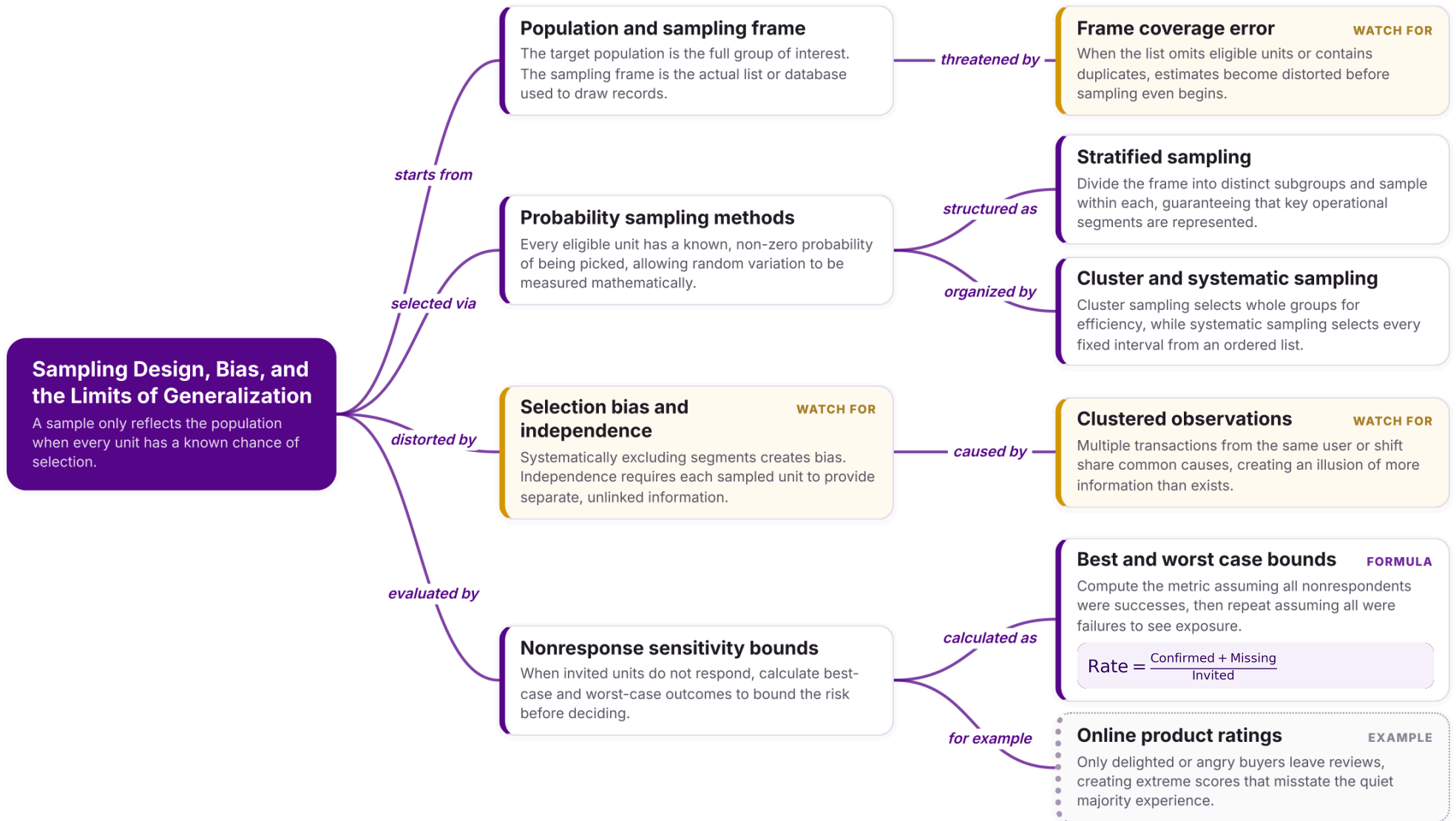
Question: How do we draw valid conclusions about business populations from sample data?

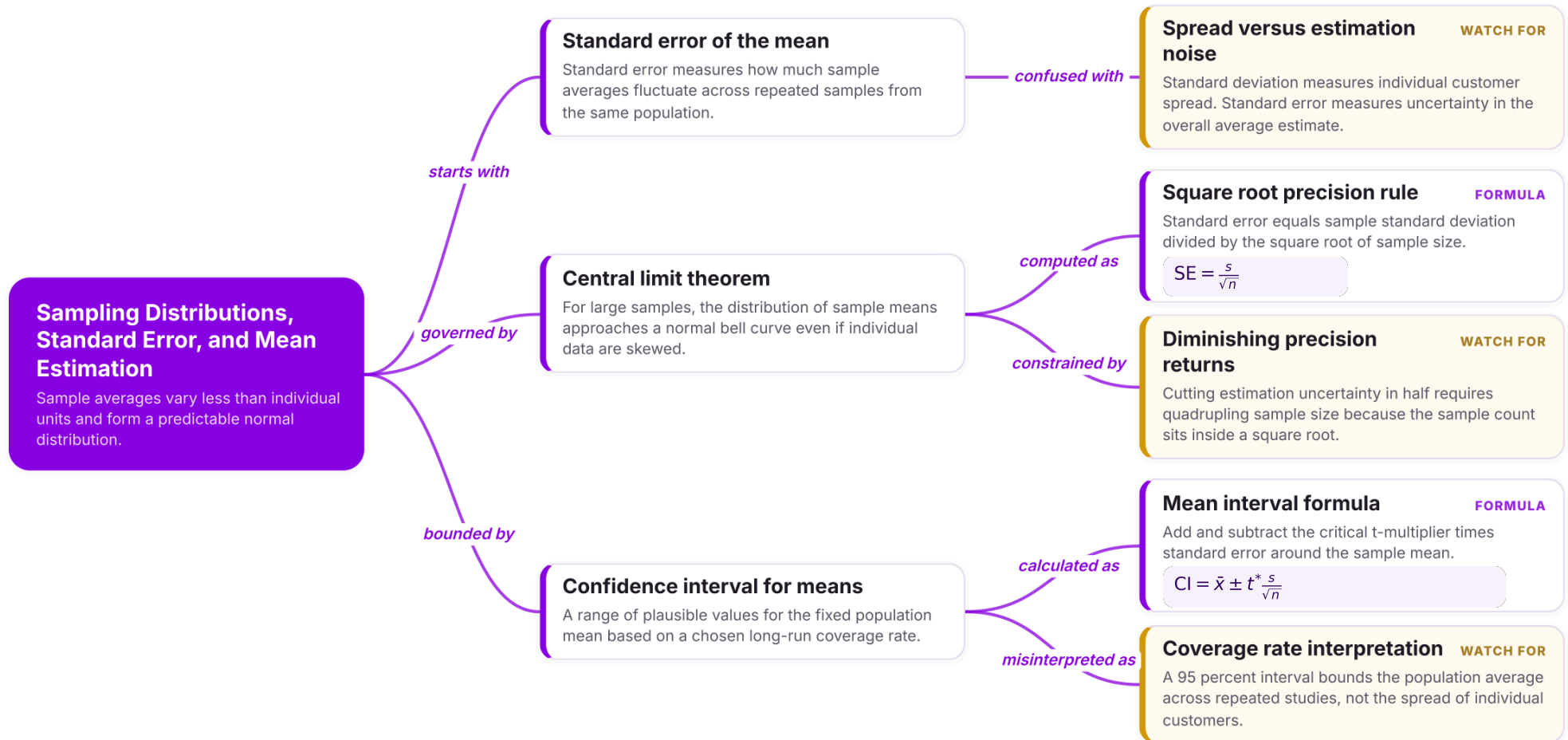


1 Nonresponse sensitivity bounds → Confidence interval for proportions: missing data widens decision uncertainty

2 Standard error of the mean → Sample size planning: error formula guides required sample

3 Probability sampling methods → Confidence interval for means: random selection enables valid intervals





Question: How do we estimate event rates and choose adequate sample sizes?